Tableau Data Analytics Project Write-up by Atal Saidy

Utilising Google Sheets and Tableau to create a professional dashboard for product sales and customer satisfaction analysis

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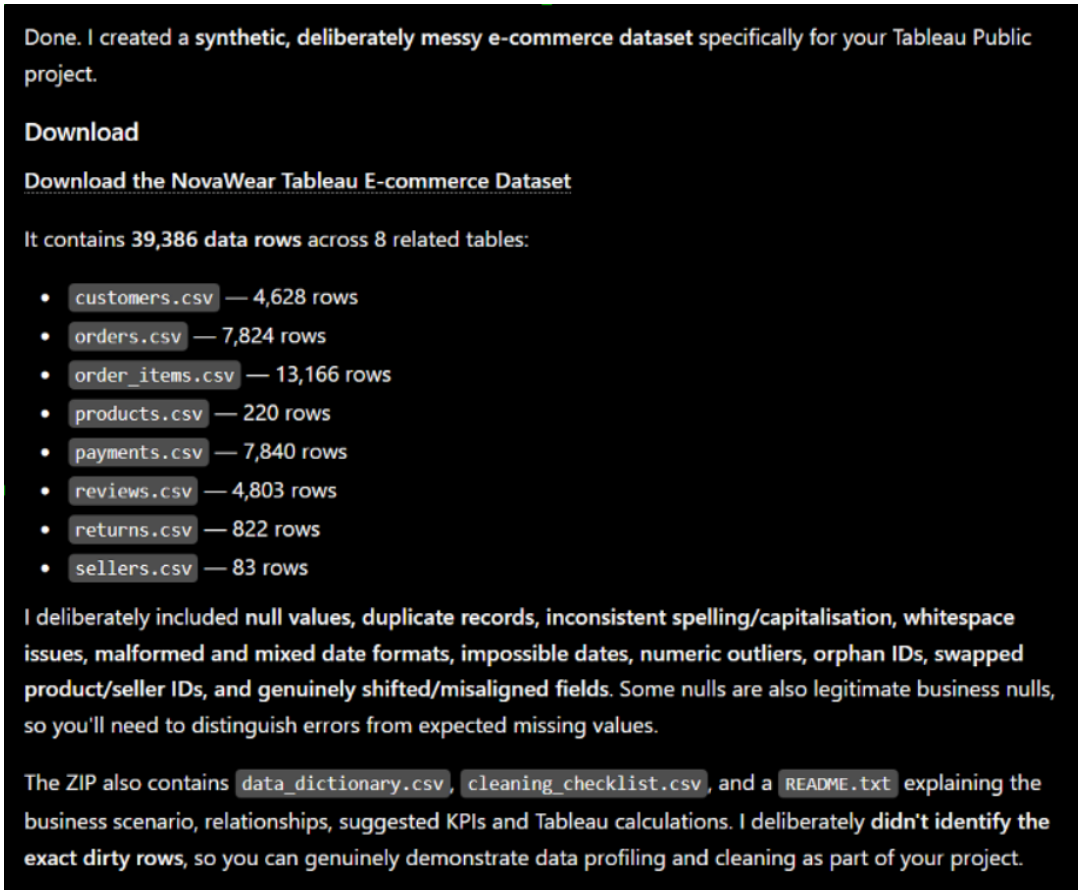
Please view my Tableau dashboard here:
https://public.tableau.com/app/profile/atal.saidy/viz/ProductSalesandCustomerSatisfaction/IntroductionPage

Project Summary

The focus of this project was to create a dashboard within Tableau for the analysis of product sales and customer satisfaction from an AI generated dataset. The main skills displayed in this project are: using formulas and built-in tools for more efficient and consistent data-cleaning, dashboard creation and familiarity with business intelligence tools, and independent research utilising AI, YouTube and official Tableau/Salesforce articles. During this project, the tools I used were: Google Sheets, due to its easy access , built-in data-cleaning and formatting tools, and ability to use formulae; NotePad++, as it is a free tool, has a more advanced find (ctrl+f) tool than regular Windows Notepad, and clear indenting for easier viewing; Tableau, due to its easy access also, option for a desktop app, and transferable skills to other Business Intelligence tools.

Data-cleaning

The datasets used in this project were generated by AI. Within my prompt, I included that the data should contain “null/misaligned values” to give me an opportunity to learn and practice data-cleaning skills. Below is a screenshot of the response from ChatGPT showing this:



Done. I created a **synthetic, deliberately messy e-commerce dataset** specifically for your Tableau Public project.

Download

Download the NovaWear Tableau E-commerce Dataset

It contains **39,386 data rows** across 8 related tables:

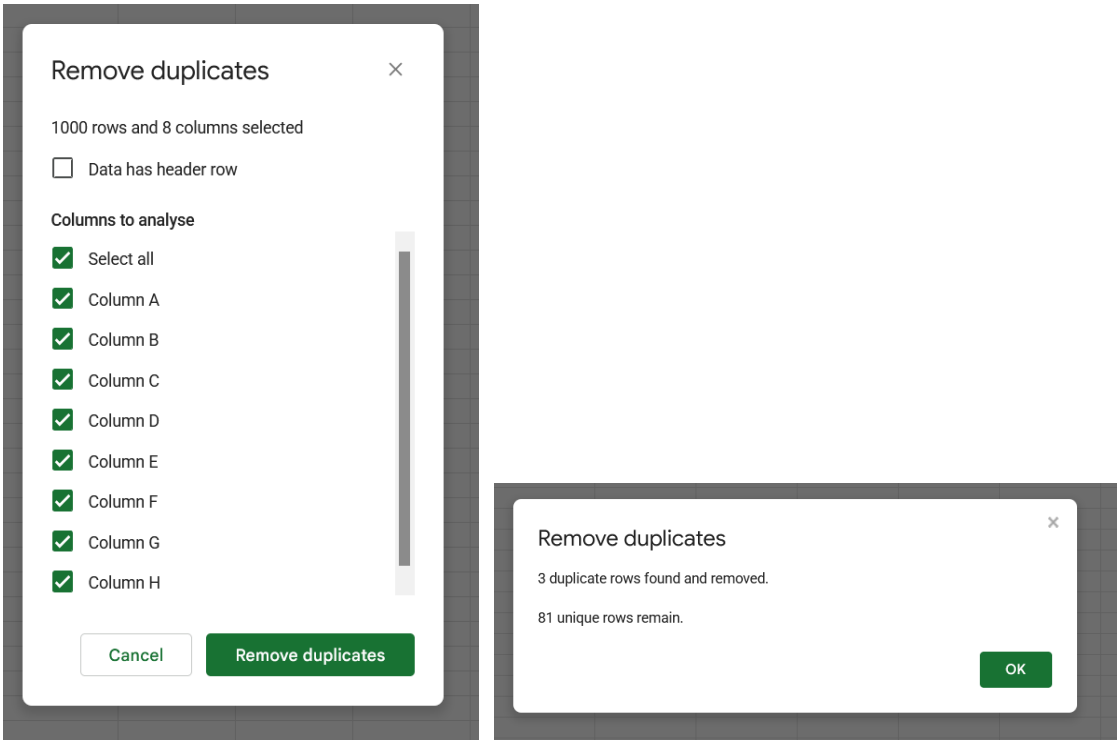
- `customers.csv` — 4,628 rows
- `orders.csv` — 7,824 rows
- `order_items.csv` — 13,166 rows
- `products.csv` — 220 rows
- `payments.csv` — 7,840 rows
- `reviews.csv` — 4,803 rows
- `returns.csv` — 822 rows
- `sellers.csv` — 83 rows

I deliberately included **null values, duplicate records, inconsistent spelling/capitalisation, whitespace issues, malformed and mixed date formats, impossible dates, numeric outliers, orphan IDs, swapped product/seller IDs, and genuinely shifted/misaligned fields**. Some nulls are also legitimate business nulls, so you'll need to distinguish errors from expected missing values.

The ZIP also contains `data_dictionary.csv`, `cleaning_checklist.csv`, and a `README.txt` explaining the business scenario, relationships, suggested KPIs and Tableau calculations. I deliberately **didn't identify the exact dirty rows**, so you can genuinely demonstrate data profiling and cleaning as part of your project.

As mentioned, I used Google sheets to complete the data-cleaning. Below is the process I followed during the data-cleaning:

1. Remove duplicates via the built-in tool (images below)



2. Identified severe outliers and either removed or adjusted these values accordingly
3. Any incorrect values which would skew analysis where marked as N/A (for example time stamps which where listed as “TBC” or “Not listed”)
4. In the case of different spellings of a term, I would use a nested IF function to convert the different spellings into the standard and moving across all other values. An example of this is when I adjusted the varying spellings for UK, the formula I used was [=IF(E2="United Kingdom", "UK", IF(E2="U.K.", "UK", E2))]. (Image below)

```
=IF(E2="United Kingdom", "UK", IF(E2="U.K.", "UK", E2))
```

B	C	D	E	F	G	H	I
seller_name	seller_cit	seller_region	seller_count	cleaned_country	seller_ra	onboarding_date	fulfilment_mod
Atlas Retail Partners 01	Manchester	England	United Kingdom	=IF(E2="United Kingdom", "UK", IF(E2="U.K.", "UK", E2))			/arehouse
Riverline Supply 02	Birmingham	England	United Kingdom	UK	4.34	2024-02-11 12:40:04	3PL
Urban Parcel Co 03	London	England	UK	UK	4.92	2022-12-07 08:20:06	3pl
MetroGoods 04	Edinburgh	Scotland	United Kingdom	UK	4.7	2022-01-04 06:35:11	3PL
Cedar Trade 05	Dublin	Leinster	Ireland	Ireland	n.a	2024-06-15 20:31:39	3PL
Atlas Retail Partners 06	London	England	United Kingdom	UK	3.84	2023-06-19 12:48:35	Owned Warehouse
Atlas Retail Partners 07	Glasgow	Scotland	United Kingdom	UK	4.3	2021-11-08 22:14:31	Owned Warehouse
GreenBridge Fulfilment 08	Birmingham	England	United Kingdom	UK	4.12	2021-06-23 02:47:42	3PL
MetroGoods 09	London	England	United Kingdom	UK	3.92	2022-07-21 21:57:49	Owned Warehouse
BlueDock Logistics 10	London	England	United Kingdom	UK	4.4	2024-01-22 23:31:45	Owned Warehouse

5. For blank values, I would use the filter option to only display the blanks and change these values to N/A instead, as mentioned this was to avoid skewing data analysis later on. (Image below)

seller	seller_name	seller_cit	seller_region	seller_count	cleaned_country	seller_ra	onboarding_date	fulfilment_mod
S0005	Cedar Trade 05	Dublin	Leinster	Ireland	Ireland	n.a	2024-06-15 20:31:39	3PL
S0047	Riverline Supply 47	Manchester	England	United Kingdom	UK	n.a	2021-04-07 13:26:42	3PL
S0053	MetroGoods 53	Birmingham	England	United Kingdom	UK	n.a	2022-07-26 05:17:00	Owned Warehouse
S0065	Riverline Supply 65	Madrid	Community of Madrid	Spain	Spain	n.a	TBC	Dropship
S0074	Urban Parcel Co 74	London	England	United Kingdom	UK	n.a	2022-02-15 22:18:19	3PL

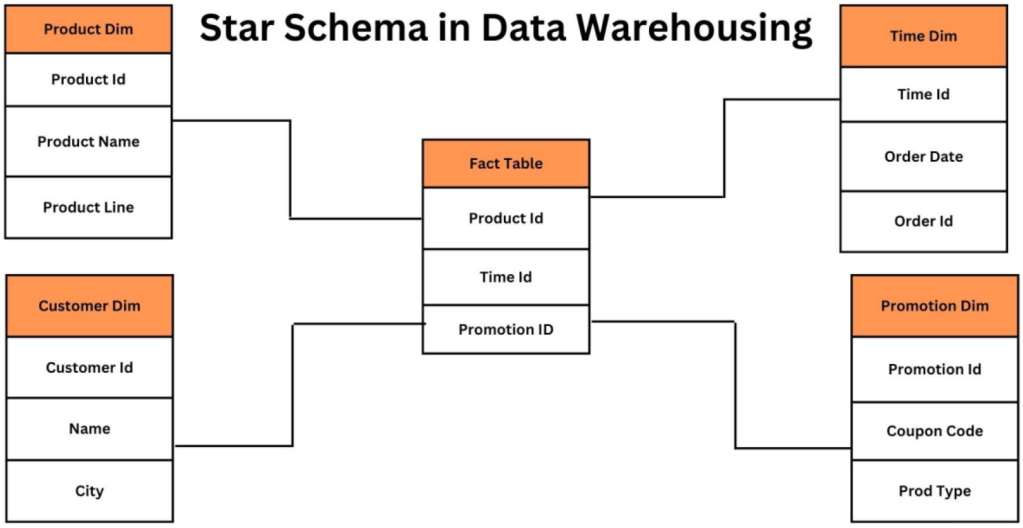
6. For Date/Time values, I would format these using the built-in Date/time formatting within Google Sheets. I practised splitting the date and time to individual columns using the SPLIT function (image below), however, this can also be done using the “Split text to columns” function under the “Data” tab in Google Sheets.

H	I	J
onboarding_date	cleaned_onboarding_date	cleaned_onboarding_time
01/05/2024 02:40:56	=SPLIT(H2, " ")	02:40:56
11/02/2024 12:40:04	11/02/2024	12:40:04
07/12/2022 08:20:06	07/12/2022	08:20:06
04/01/2022 06:35:11	04/01/2022	06:35:11
15/06/2024 20:31:39	15/06/2024	20:31:39
19/06/2023 12:48:35	19/06/2023	12:48:35
08/11/2021 22:14:31	08/11/2021	22:14:31
23/06/2021 02:47:42	23/06/2021	02:47:42

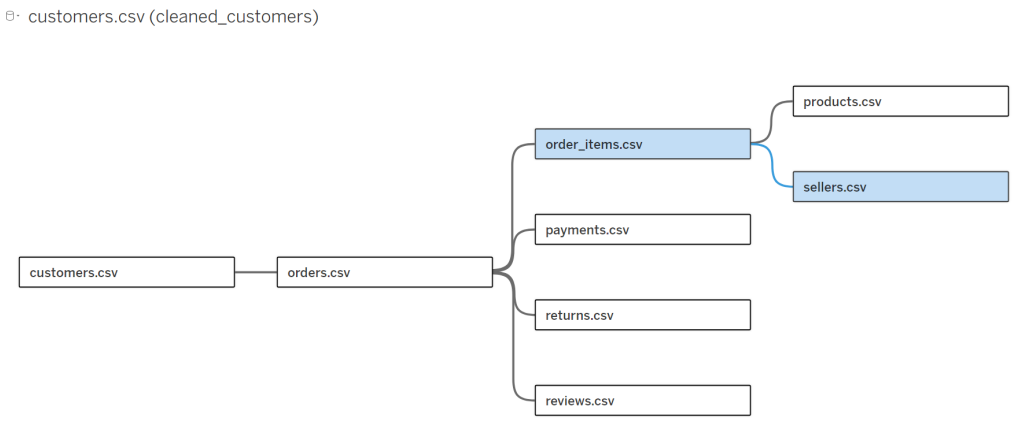
These steps were repeated across the datasets.

Dashboard Creation

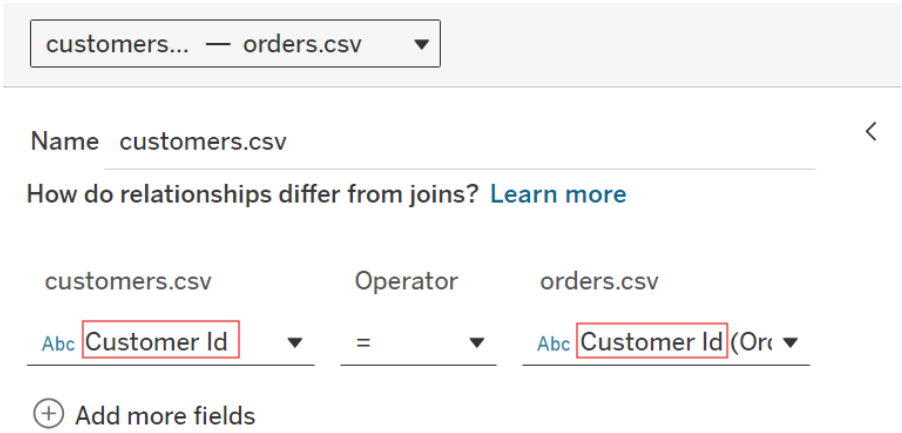
I started by importing the now clean data into my new Tableau project within the Desktop app. I then created the data model for this dashboard following a snowflake schema. The common alternative schema is star schema which there is an image of below:



Below is an image of the data map I created in Tableau to link my datasets.



Each of the connections between each dataset is linked based on a primary/foreign key system, which in this case, was a combination of IDs (Order, Product, Seller, Customer, etc.).Image below showing an example connection:



During the creation of the individual worksheets and dashboards, I learned: how to implement different types of charts within Tableau (vertical/horizontal bar charts, stacked bar charts and radial graphs); the importance of using a worksheet to create the visualisation and then creating the correct layout of those sheets within the dashboards; how to setup navigation objects within dashboards for improved user viewing; and data manipulation within Tableau (creating groups within datasets).

I have included an explanation of each dashboard page and chart for easier viewing below.

Dashboard explanation:

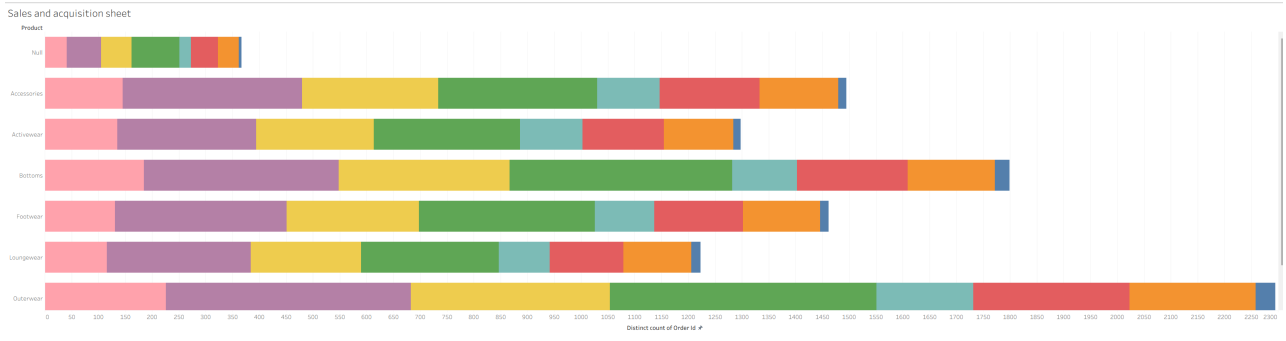
Introduction:

Provides an introduction to the dashboard and provides background on the project. Navigation buttons at the bottom of page for quick access to Sales Overview and Customer Feedback Analysis.

Sales Overview:

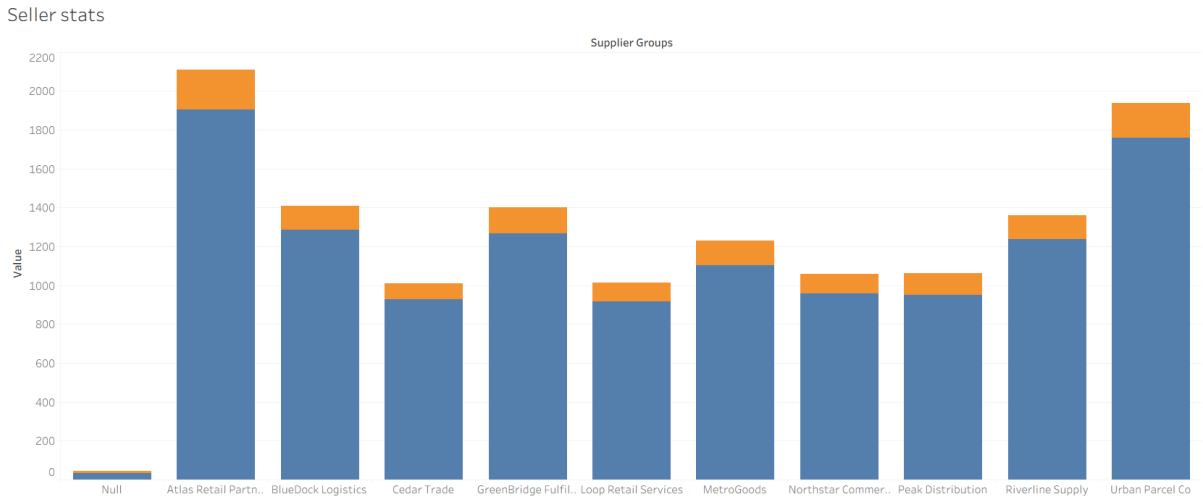
Sales and acquisition sheet:

Stacked horizontal bar chart. Shows quantity of orders per product category. The different shades in each bar show the acquisition channel used for the sale.



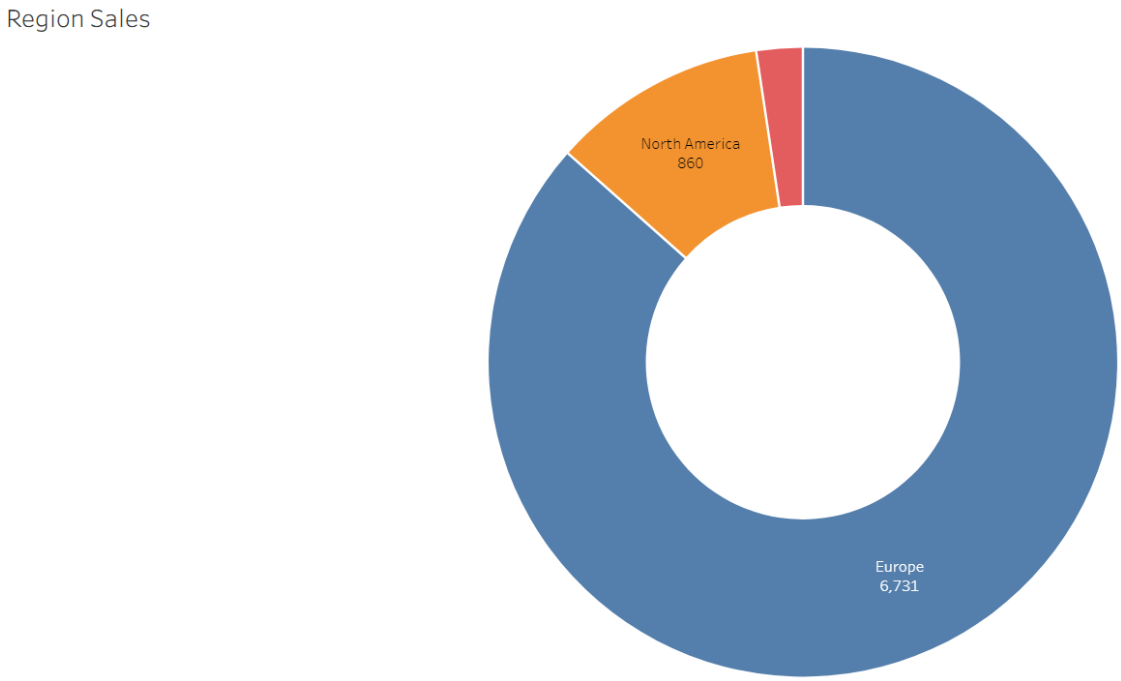
Seller stats:

Stacked bar chart. Shows number of orders (blue) and returns (orange) per supplier group.



Region Sales:

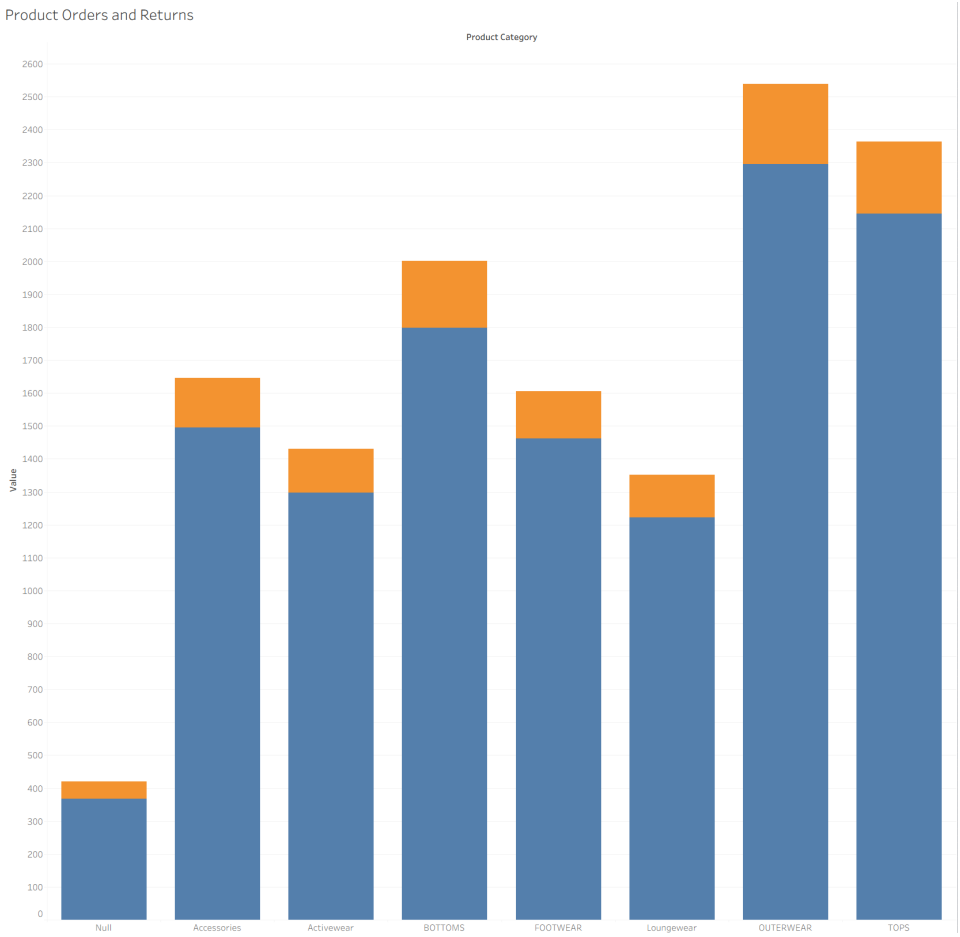
Radial chart. Showing sales per region group.



Customer Feedback Analysis:

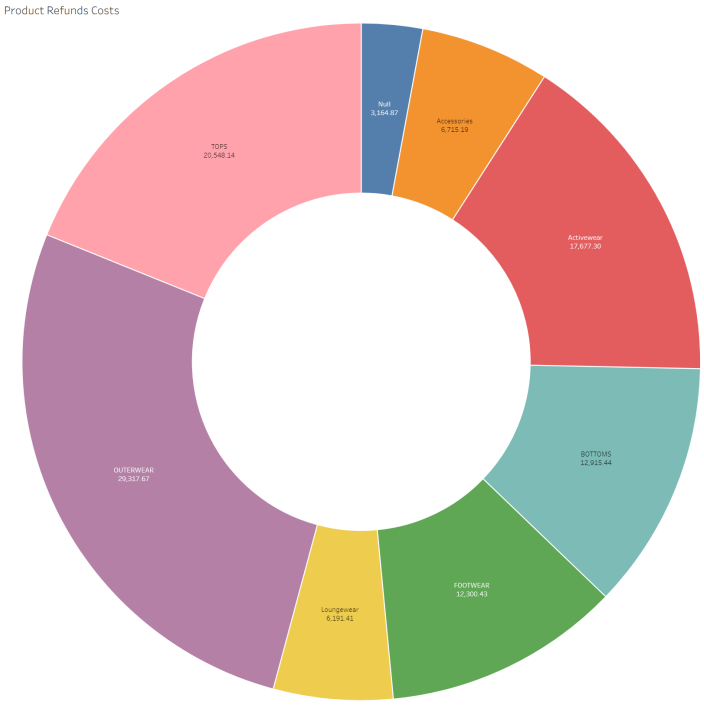
Product Orders and Returns:

Stacked bar chart. Shows orders (blue) and returns (orange) quantity per product group.



Product Refund Costs:

Radial chart. Shows total refund costs in GBP per product group.



Project Conclusion

During this project I have demonstrated and developed upon skills in data cleaning (Google Sheets), dashboard creation (Tableau) and independent research (Official vendor articles/documentation and AI).